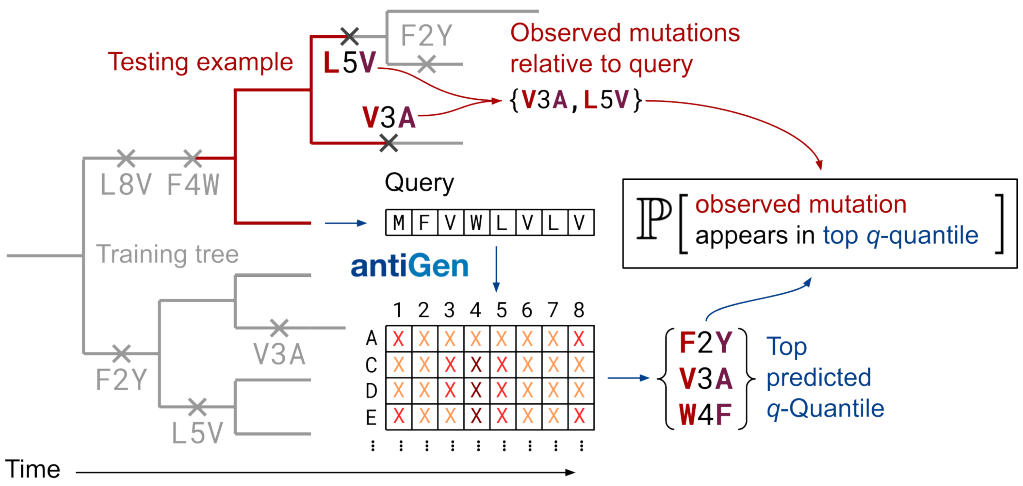
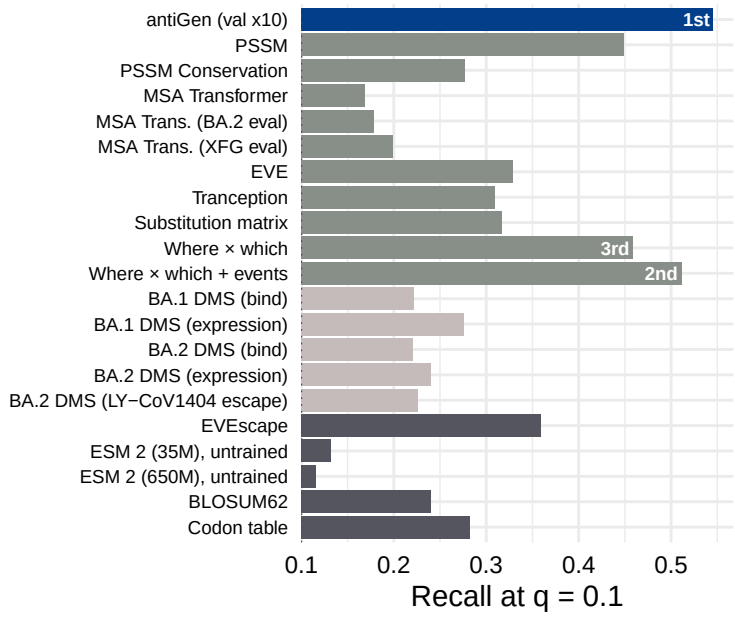
Model type Deep Mutational Scans Pandemic Data in Training Window Pre−Pandemic Data Only Val−selected (10x)

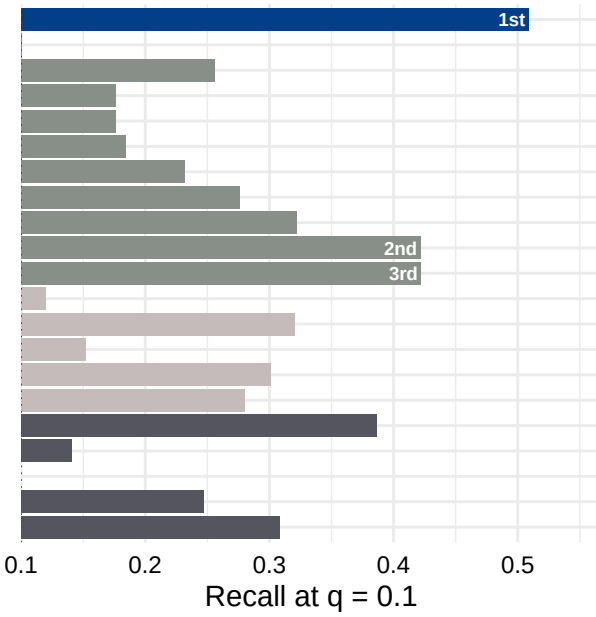
A Definition of next−mutation recall



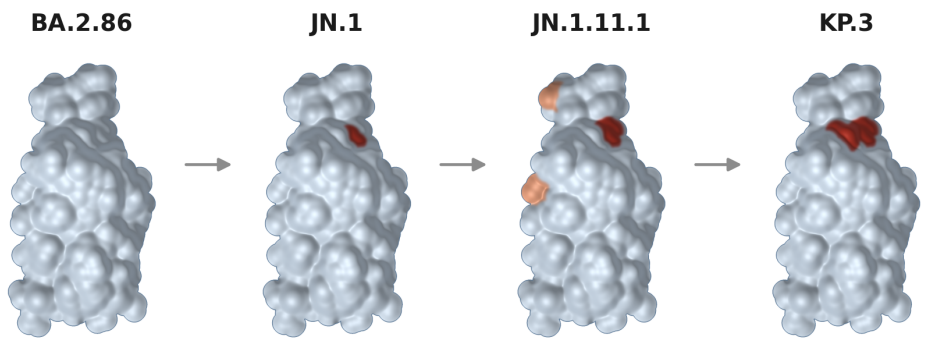
B Top decile recall, all escape mutations



C Top decile recall, de novo escape mutations



D Case study: BA.2.86 to KP.3

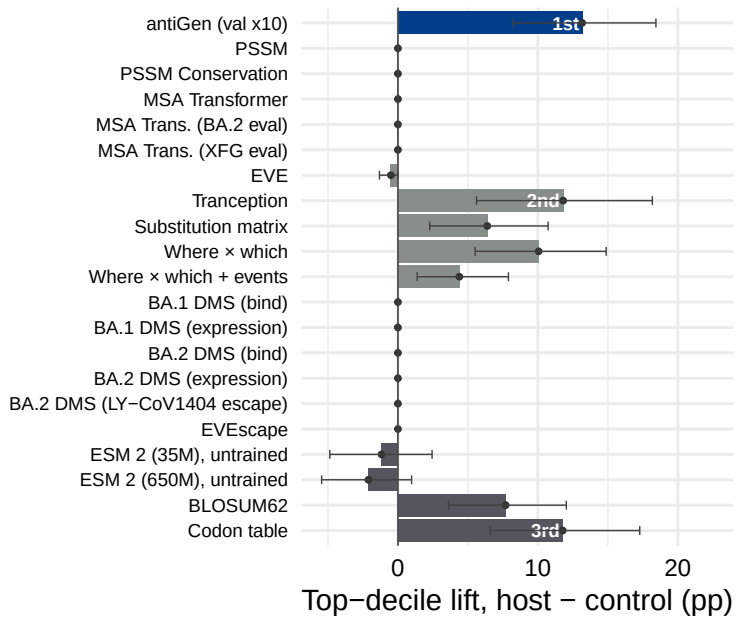


L455S 258 **V483D** 1,107 **D483V** 261
R346T 233 **T346R** 19
F456L 82 **Q493E** 191

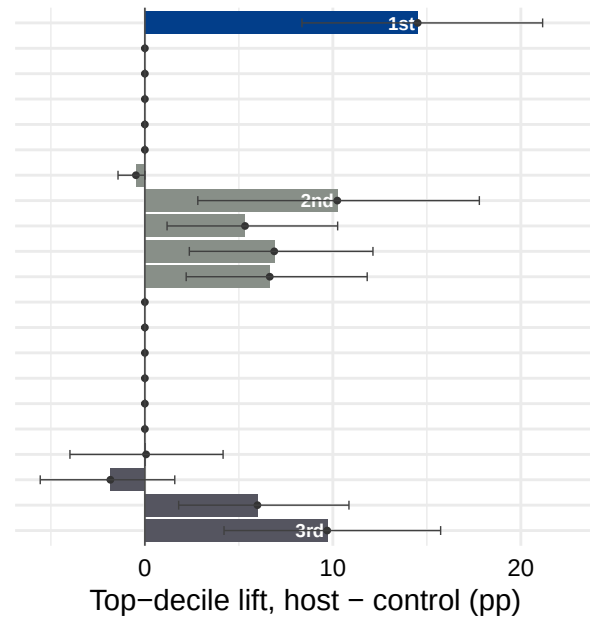
number beside each change: antiGen's rank among all 3,819 RBD substitutions

still changed in KP.3 reverts along the way

E Query adaptability, all escape mutatio



F Query adaptability, de novo escape mutations



G DMS correlation

Model type Val−selected (10x) Pandemic Data in Training Window Pre−Pandemic Data Only
Transformation Mean Mean, log None Log

H Top decile recall, all mutations